

Sandwich Insulated Panels with Textile-reinforced Concrete Skin and Aerated Concrete Core

Version 1a

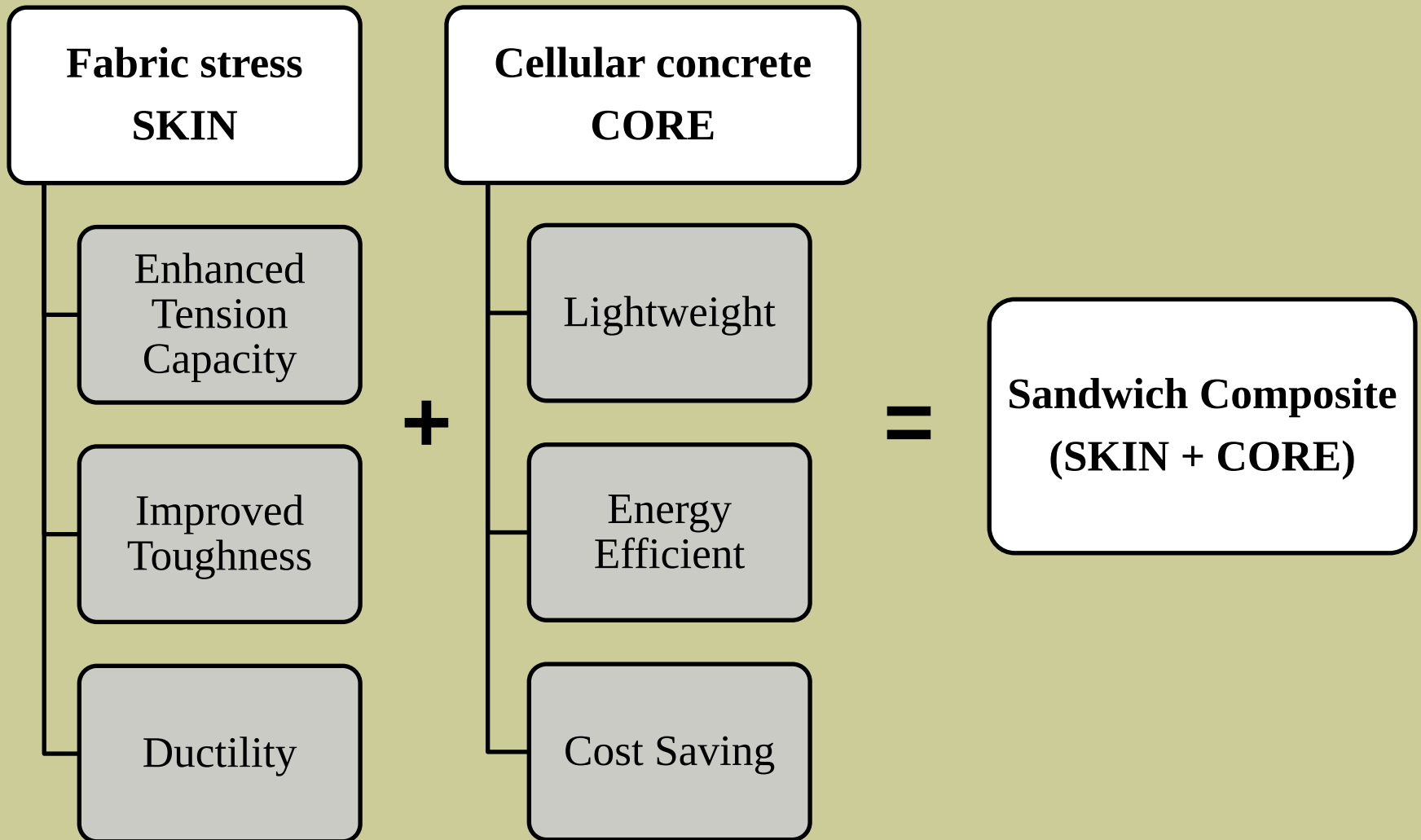
June 2013

Existing innovative systems

- o Structural Insulated Panels (SIP)
- o Solid Foam Core
- o Laminated Plywood Skins
- o Cost Savings in heating/cooling
- o Fast Construction

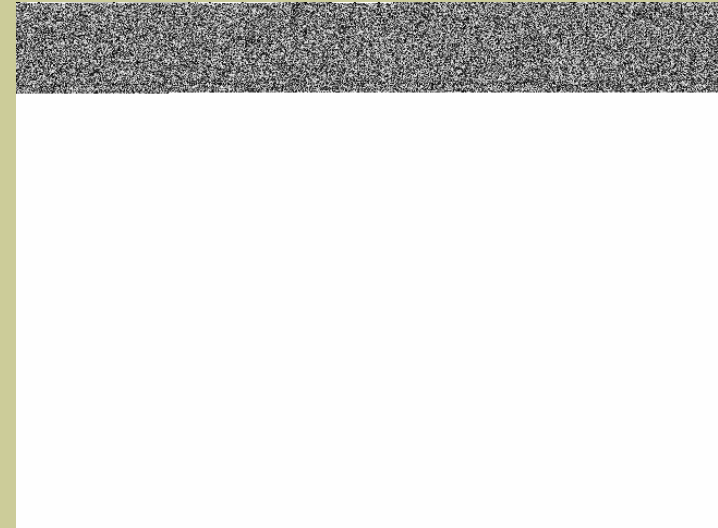


Proposed Construction system



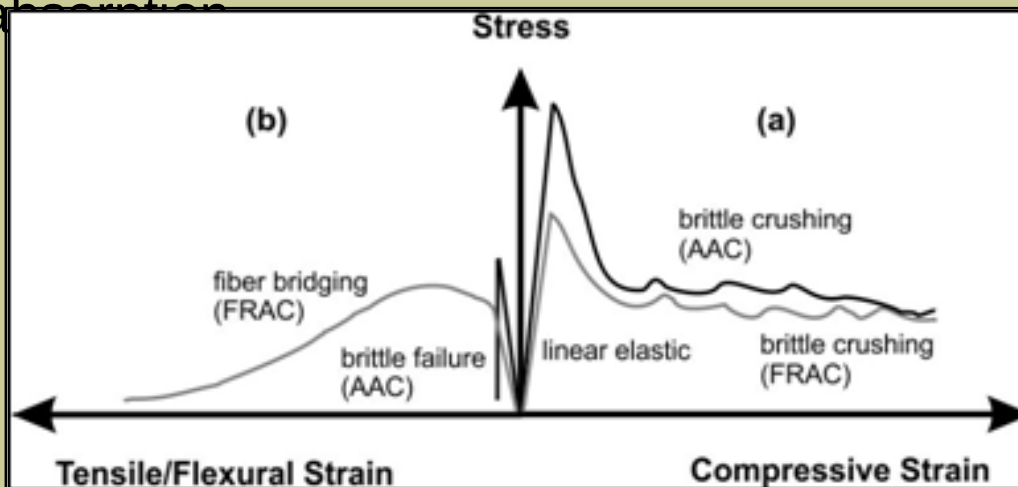
Aerated Concrete

- ❑ Aerated Concrete (AC) is a lightweight, noncombustible cement based material with high fly ash substitution.
- ❑ If AC is autoclaved for accelerated strength gain - **Autoclaved Aerated Concrete (AAC)**
- ❑ If AC is reinforced with short polymeric fibers (such as polypropylene) - **Fiber Reinforced Aerated Concrete (FRAC)**.



FRAC – Highlights

- ❑ Discontinuous porous structure
- ❑ Good thermal conductivity
- ❑ Low density, low compressive strength
- ❑ Addition of fibers improves the energy absorption



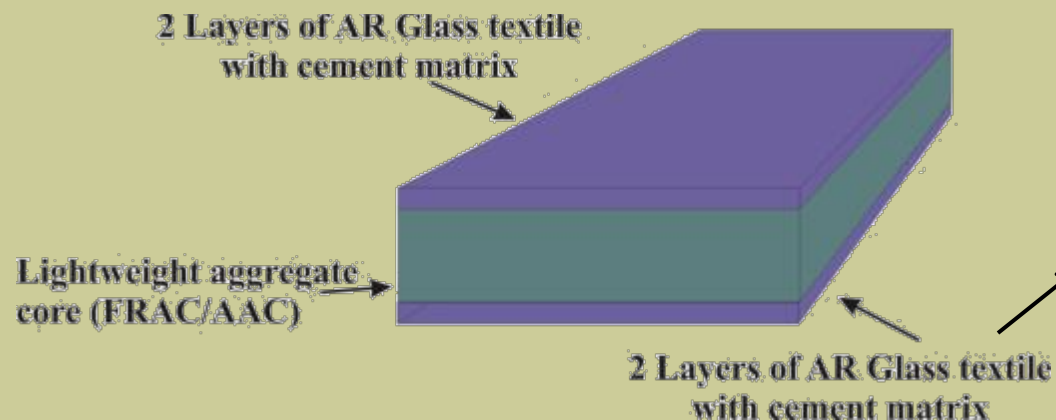
Applications :

- ❑ Load bearing and non-load bearing structures - commercial, industrial and residential applications.
- ❑ Masonry blocks; Wall cladding, Floor and Roof Panels

Sandwich Composites with Aerated Concrete

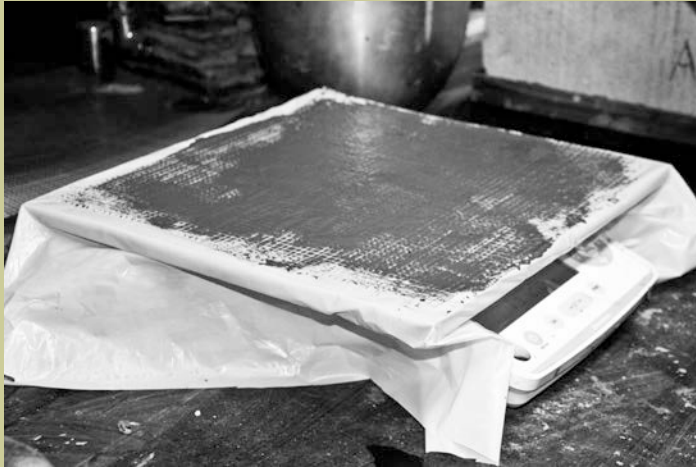
- ❑ Novel composite typically used in defense structures, exposed to shock loading
- ❑ Bonded AR glass fabric consisting of perpendicular yarns (warp and weft).
- ❑ Filament diameter = $13.5\ \mu\text{m}$, Tensile strength = 1270 – 2450 MPa, and Elastic modulus = 78 GPa.
- ❑ 2 layers of AR glass textile used as top and bottom reinforcement with cement matrix

Mix Design of Cementitious Matrix	
Constituents:	Weight (in grams)
Portland Cement	900
Water	180
Fly Ash	310
Super plasticizer	2



Sandwich Fabrication Process

Stage 1



Stage 2



Stage 3



Stage 4

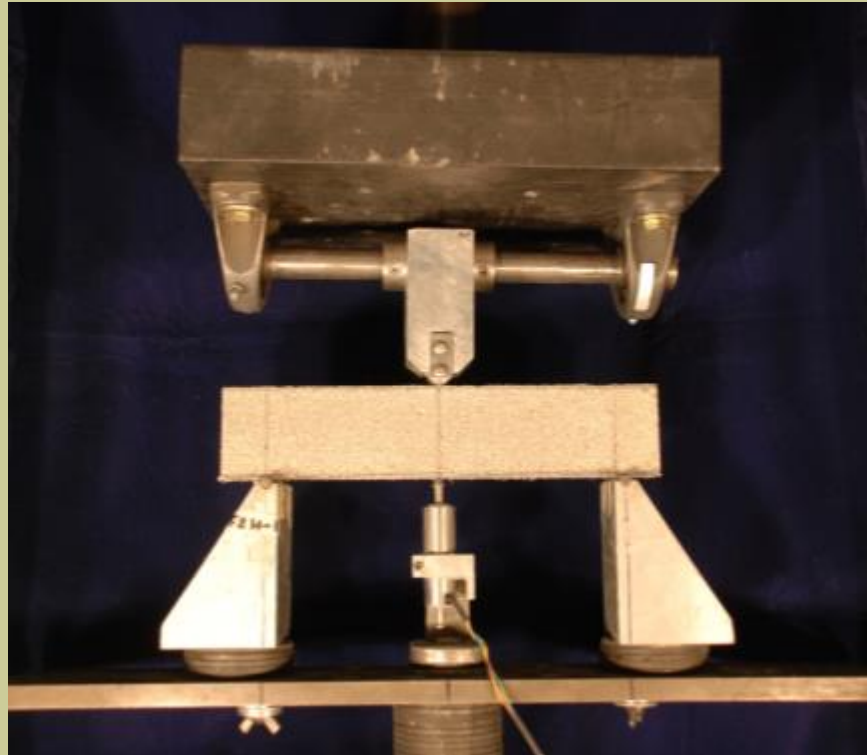


Sandwich Composites - Highlights

- ❑ Improved structural strength via enhanced tension capacity, increase in flexural strength and higher ductility
- ❑ Distributed cracking mechanism, strain hardening behavior
- ❑ Ease of manufacturing, excellent anchorage and bond development
- ❑ Distributed textile reinforcement and proper bonding with cementitious matrix

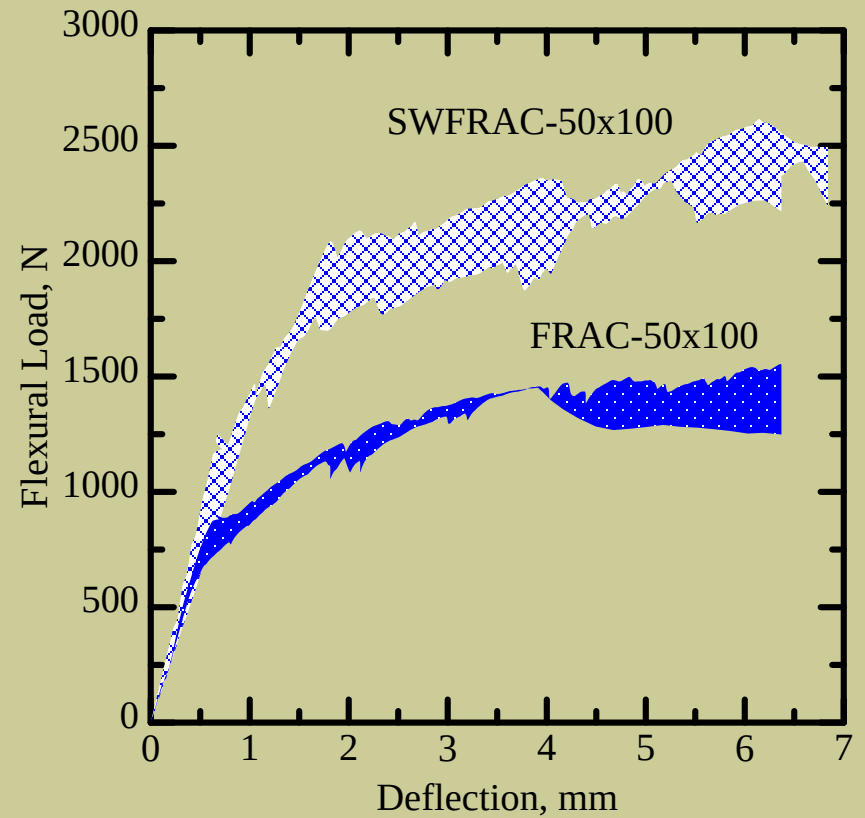
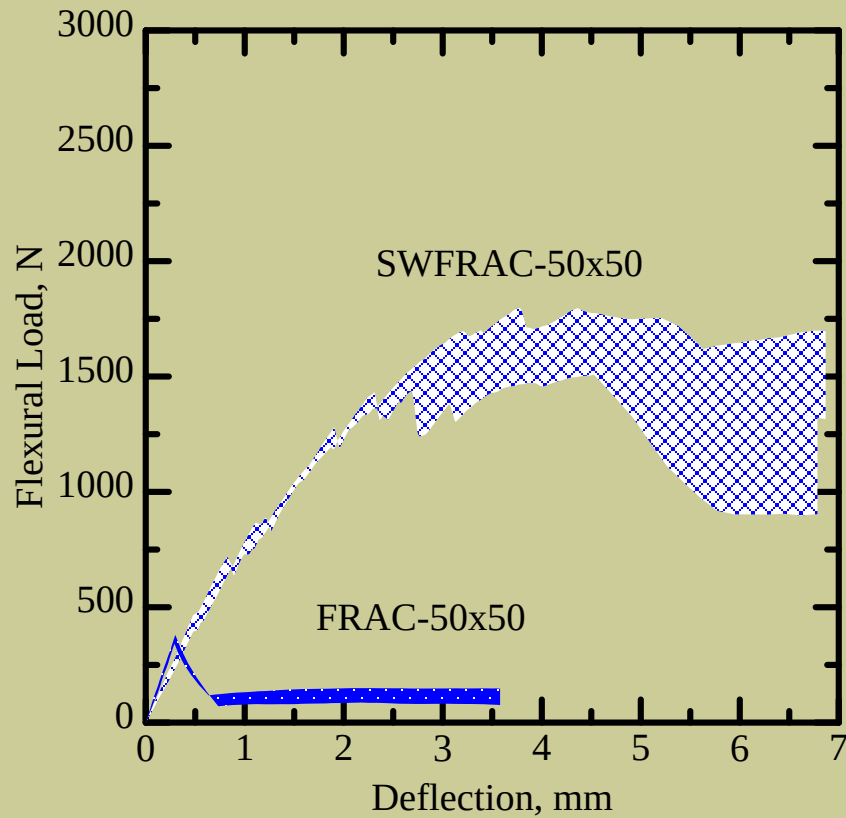


Static Flexural Test

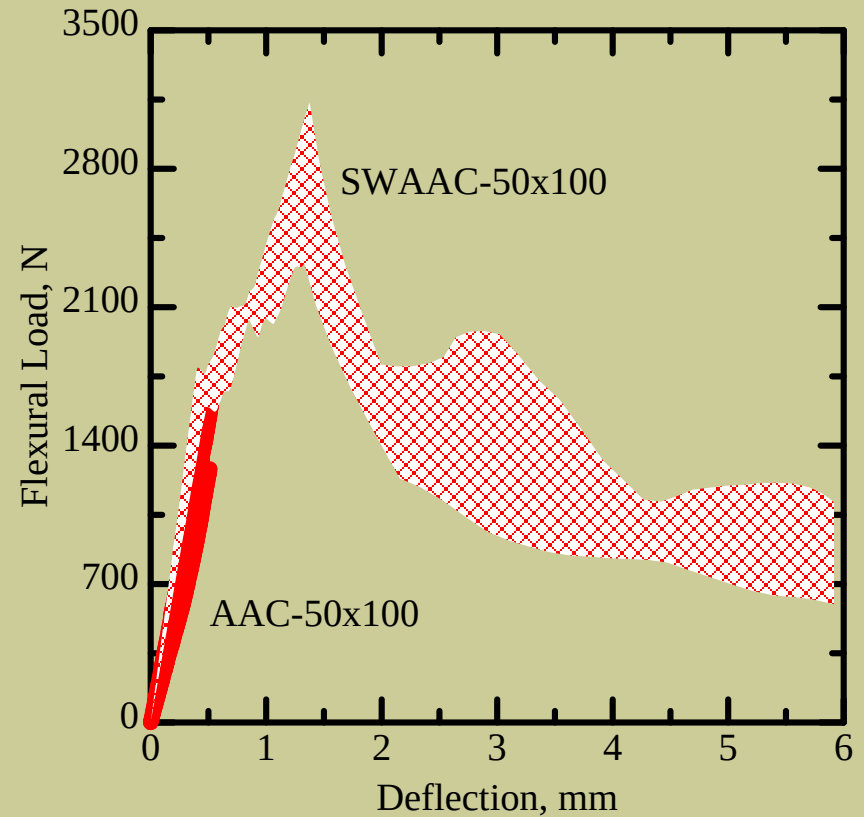
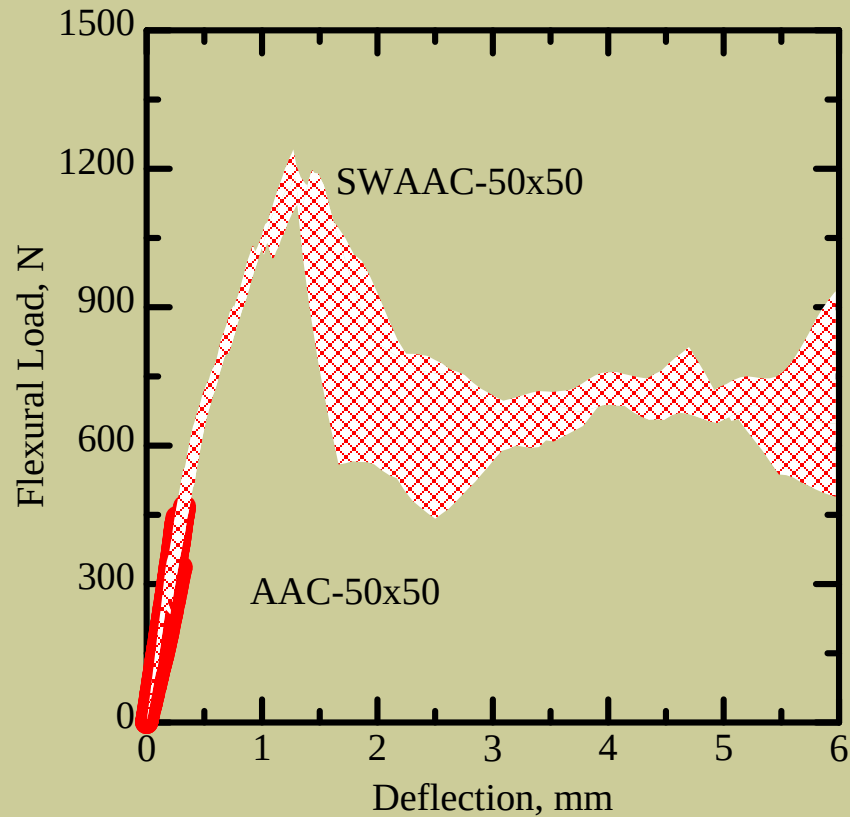


- ❑ 3 – Point Bending test under Displacement Control
- ❑ Rate of loading – 0.015 in/min (Core); 0.05 in/min (Sandwich)
- ❑ Length of specimen : 250 mm; Span : 200 mm (Constant)
- ❑ Cross-section of specimen : 50 x 25; 50 x 50; 50 x 100 mm

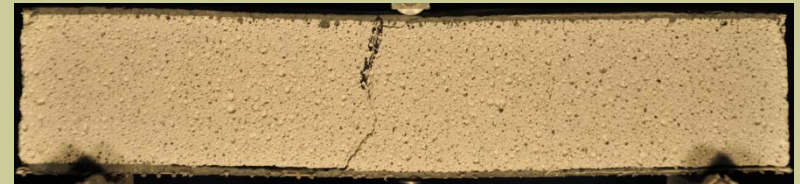
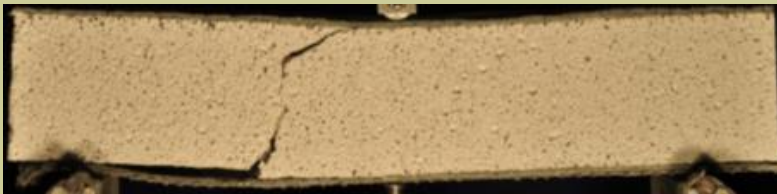
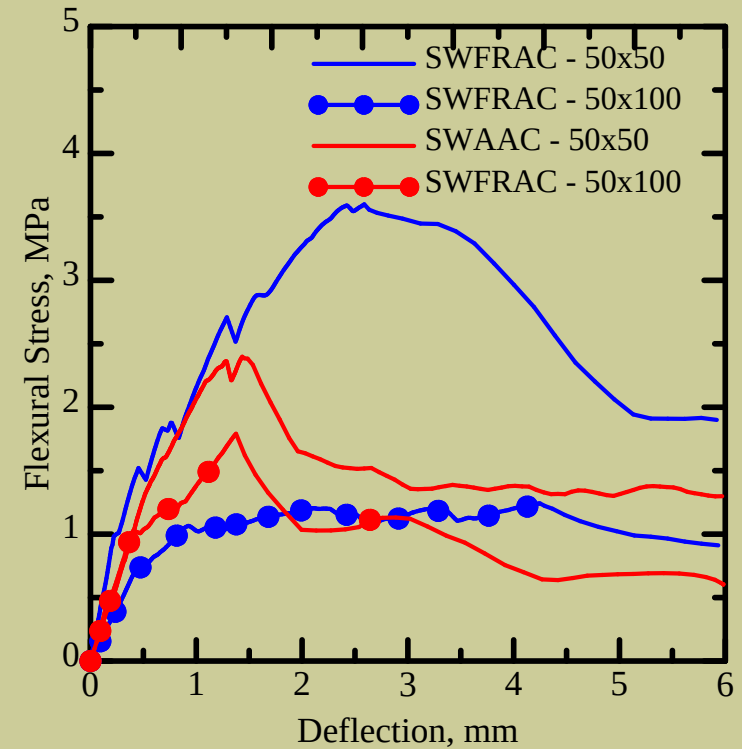
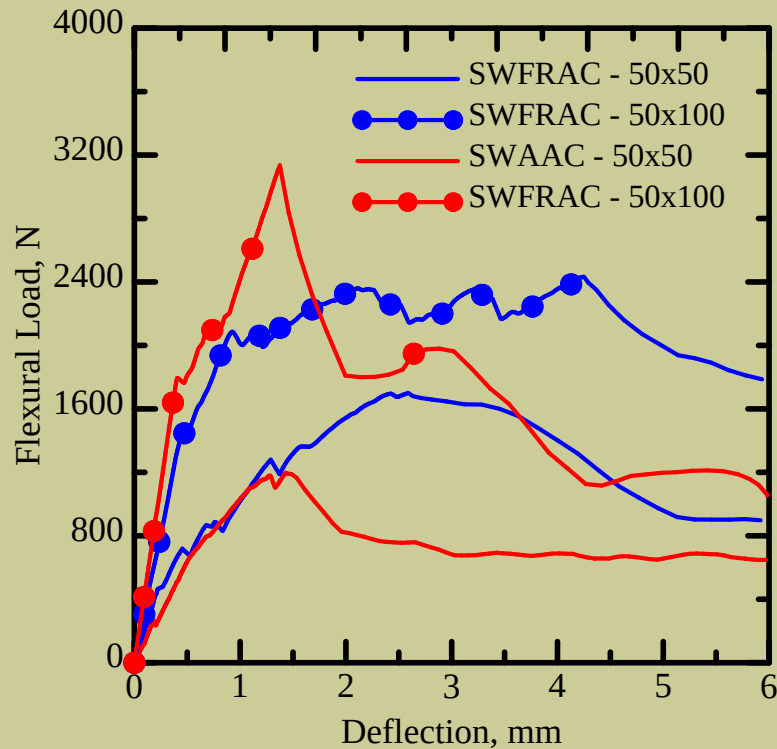
FRAC – Core v/s Sandwich- size effect



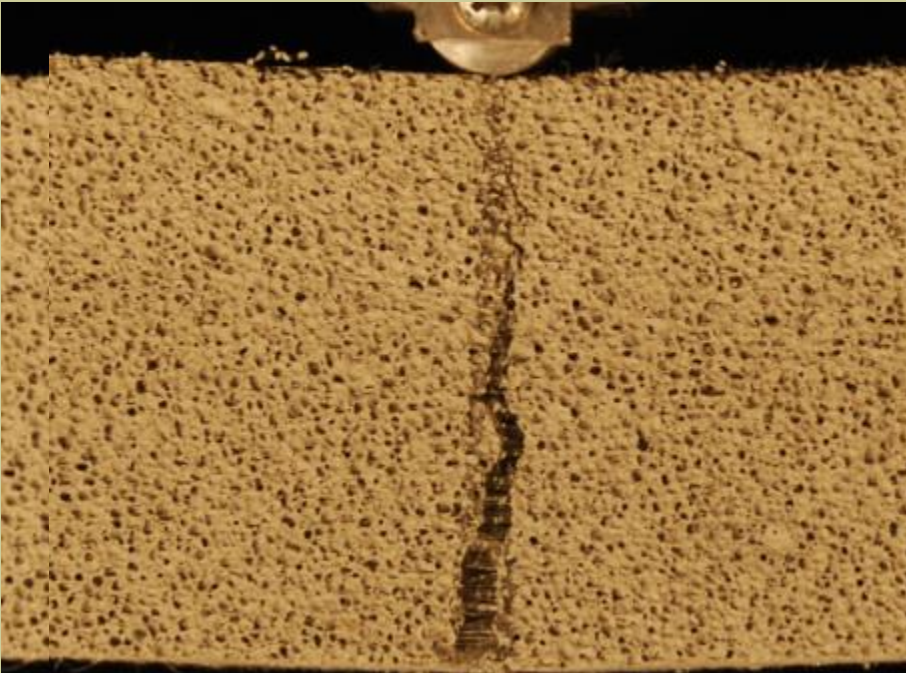
AAC – Core v/s Sandwich



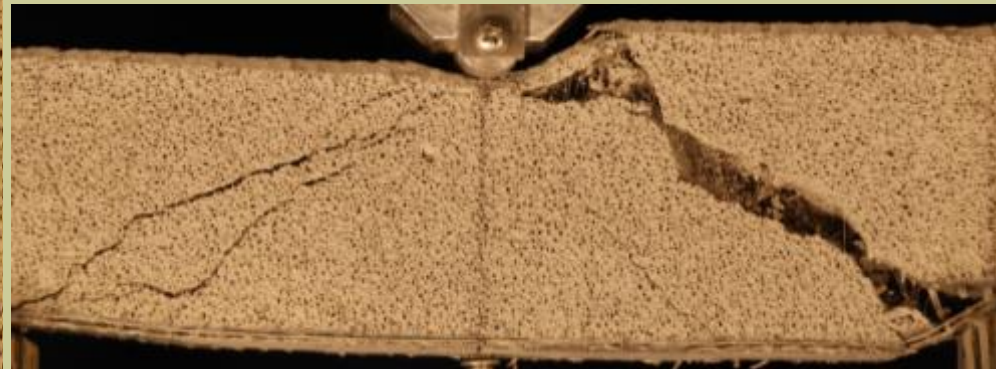
Sandwich – FRAC v/s AAC



Cracking Mechanism in Sandwich with FRAC

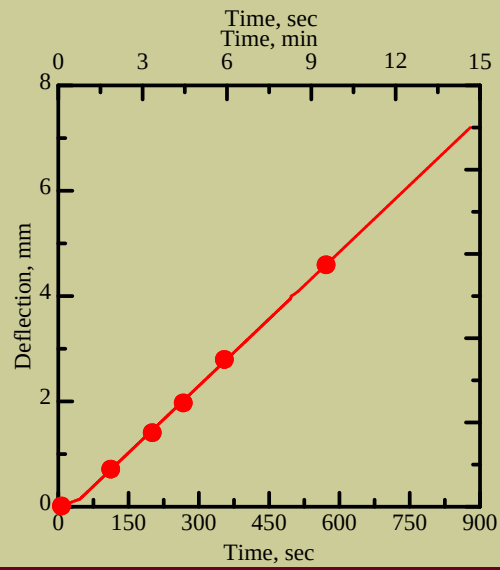
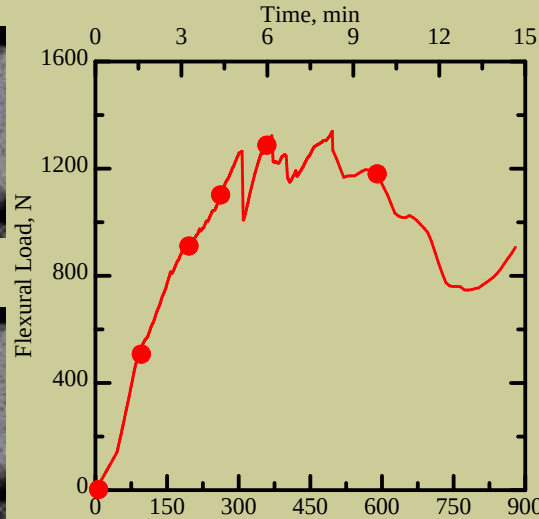
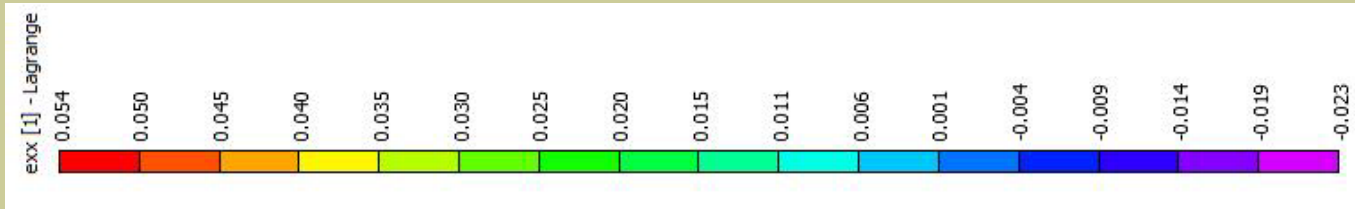
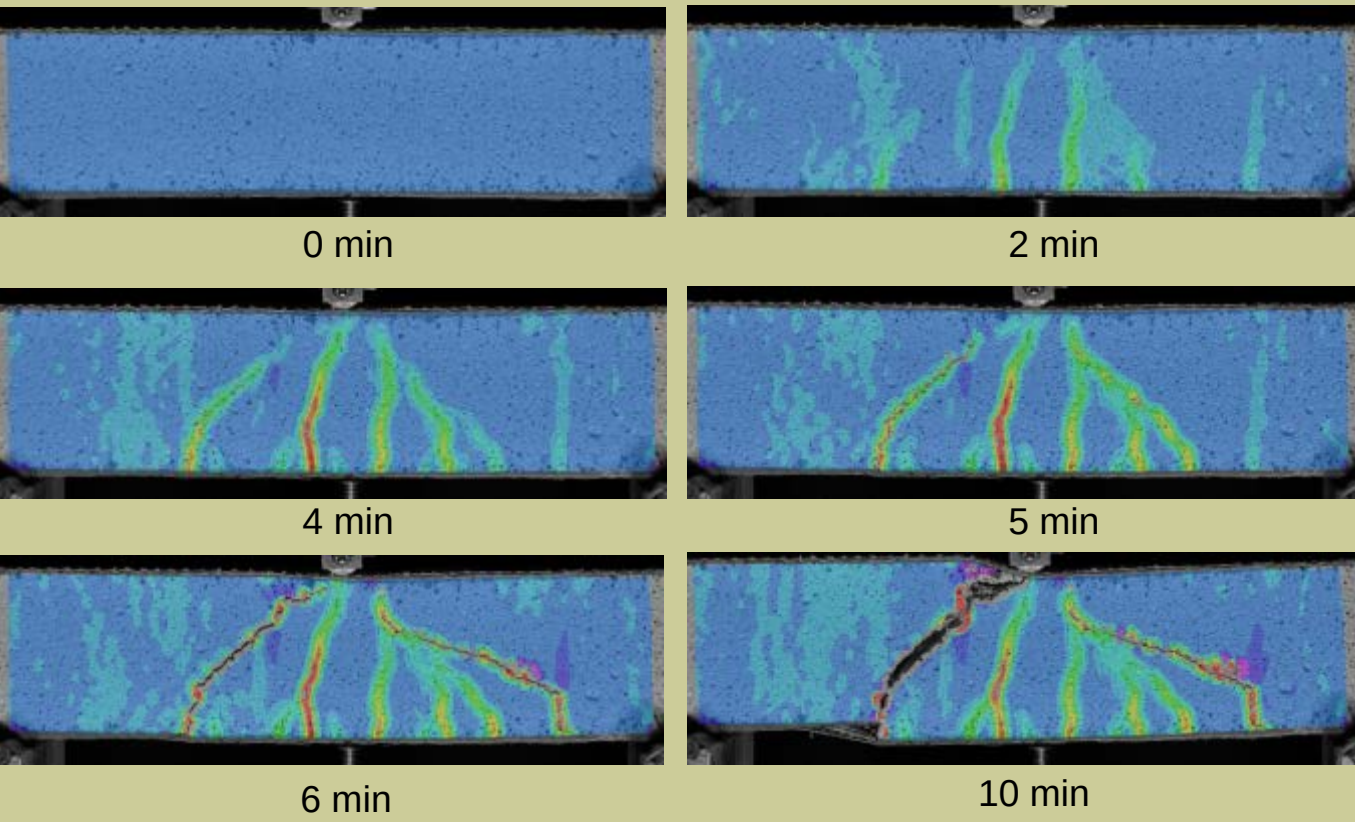


~~Time~~ = 14 mins
~~Load~~ = 220 N
Displacement = 4.5 mm
Flex. Stress = 0.5 MPa

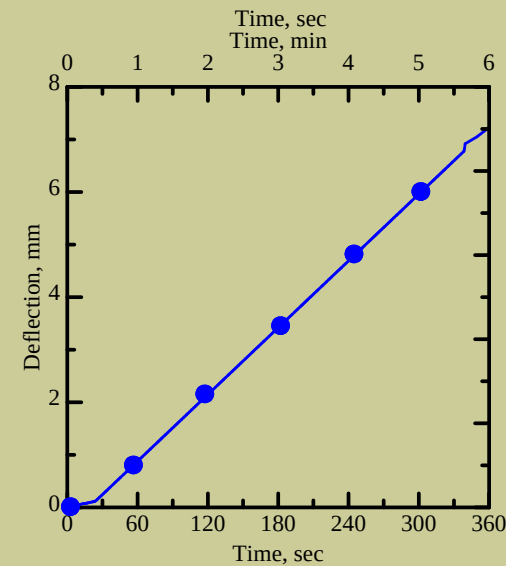
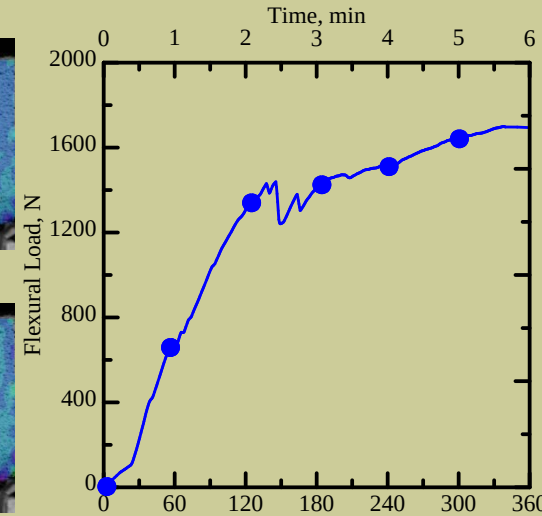
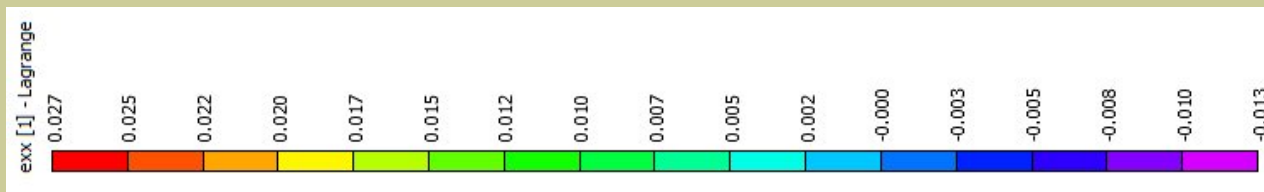
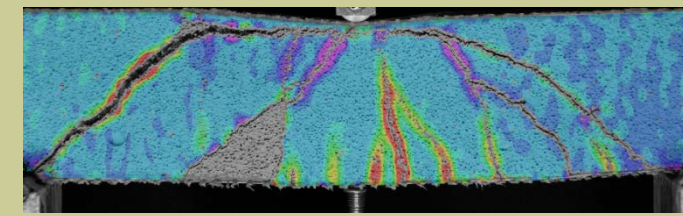
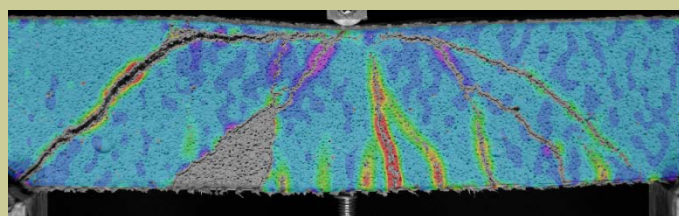
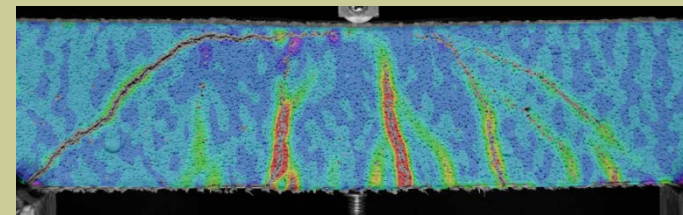
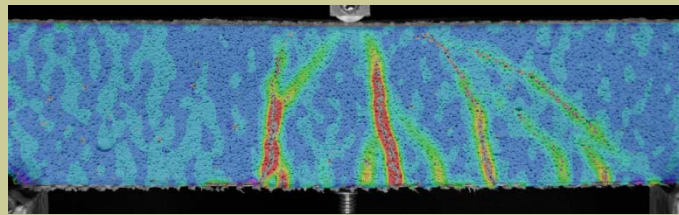
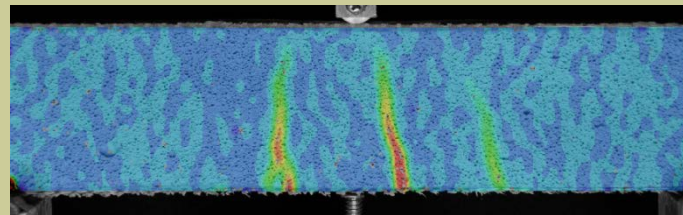
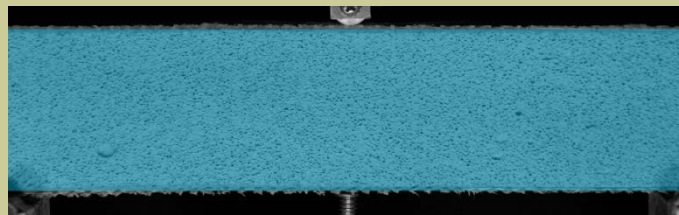


~~Time~~ = 10 mins
~~Load~~ = 220 N
Displacement = 9 mm
Flex. Stress = 2.0 MPa

AAC Sandwich



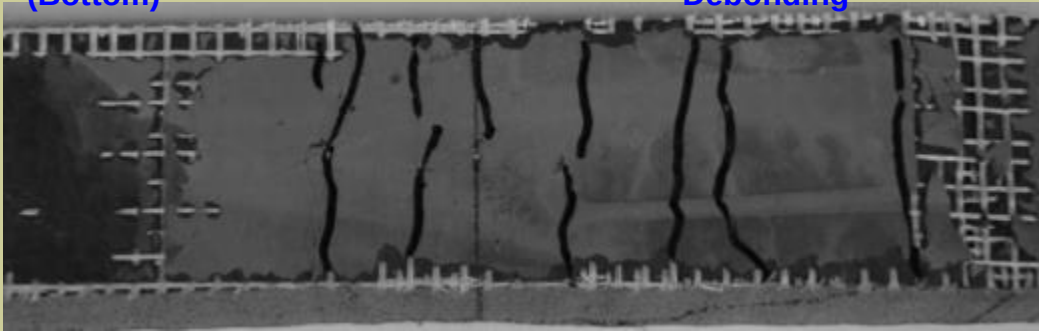
FRAC Sandwich



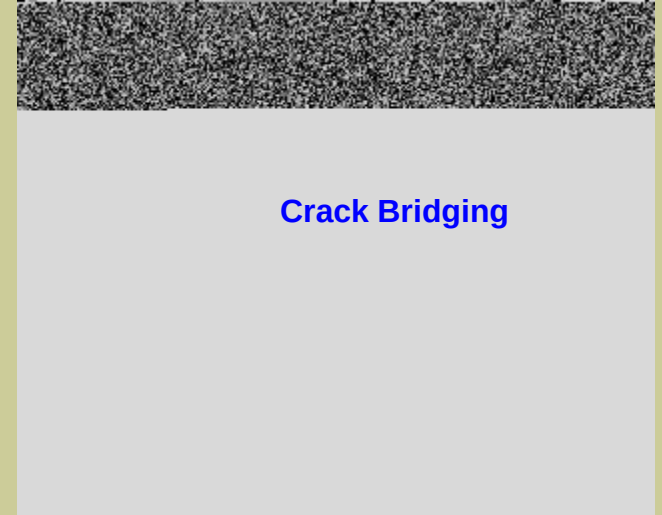
Distributed Cracking Mechanism in TRC Sandwich Composites - Static Loading

Flexural Cracks
(Bottom)

Textile-Matrix
Debonding



Crack Bridging

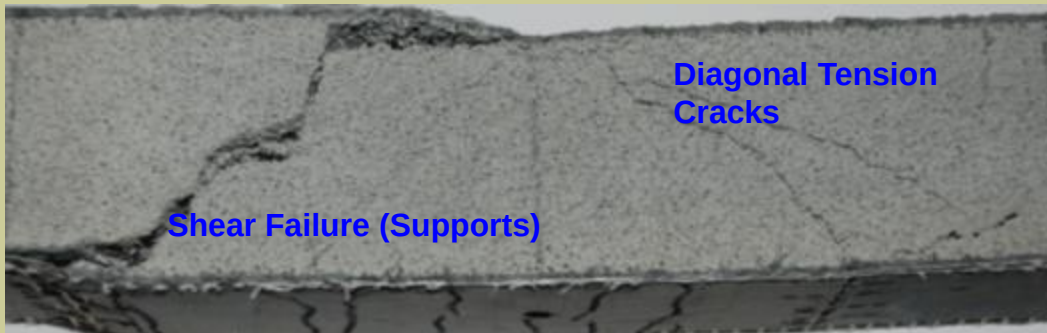


Buckling of TRC (Top)

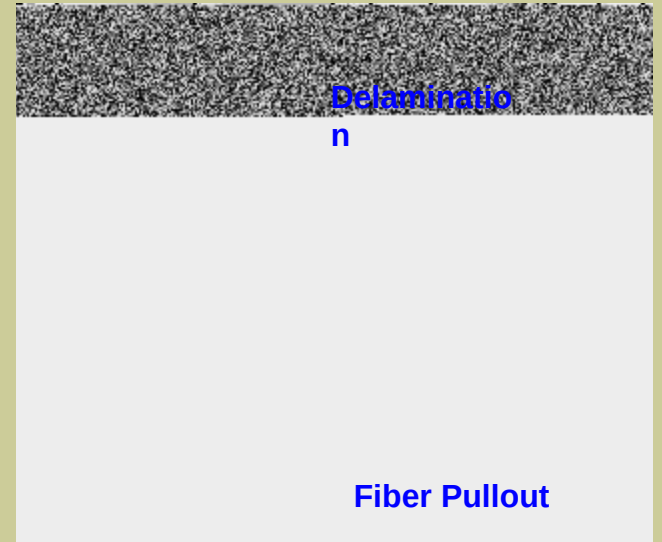


Diagonal Tension
Cracks

Shear Failure (Supports)



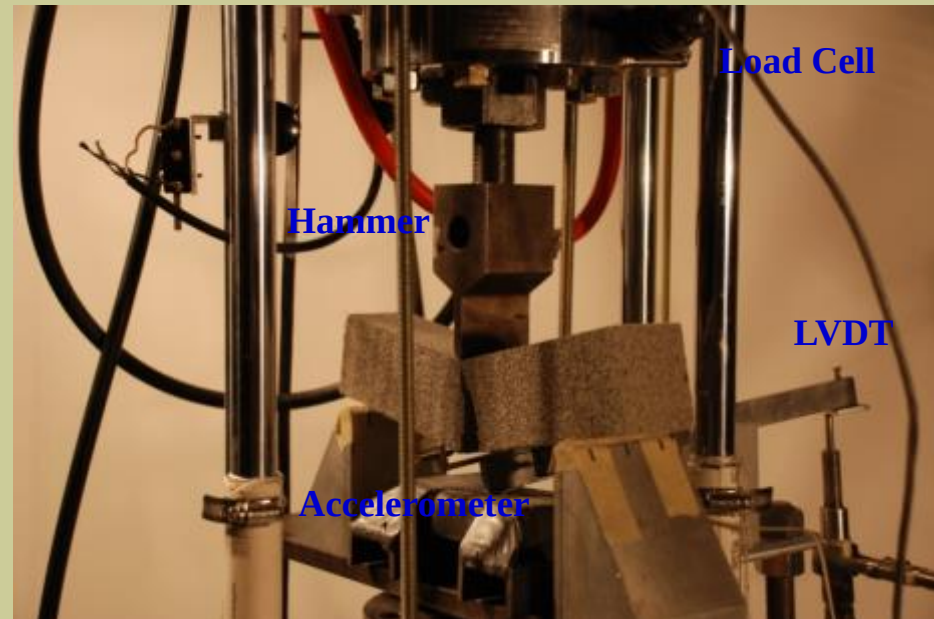
Delamination



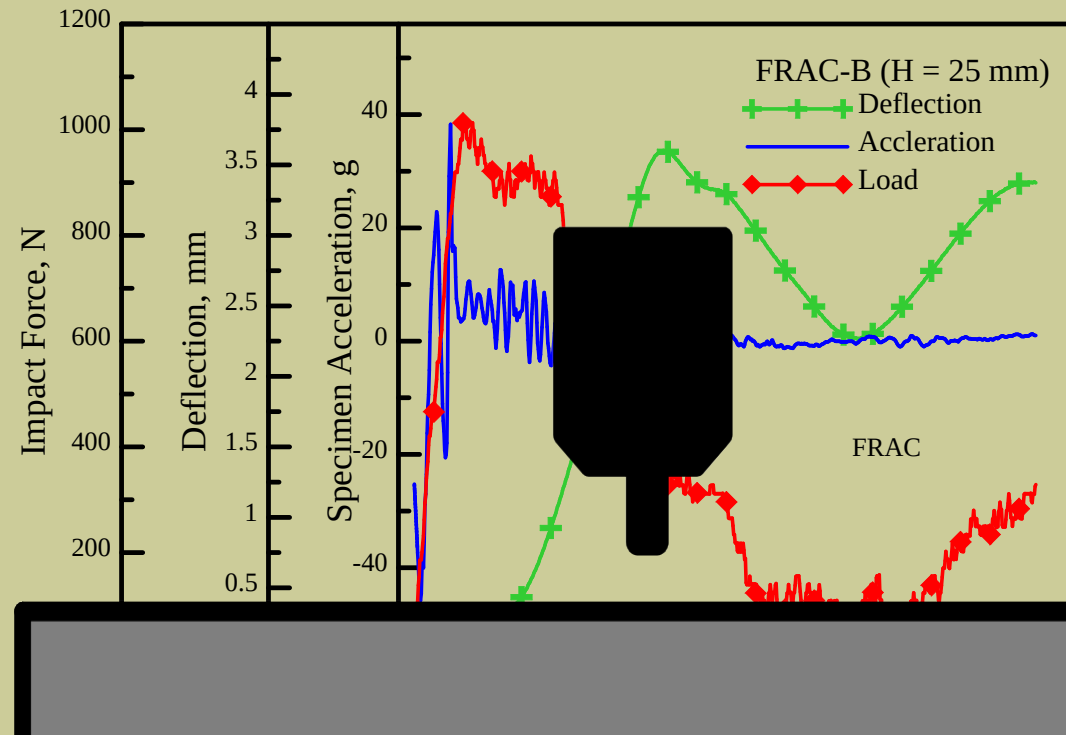
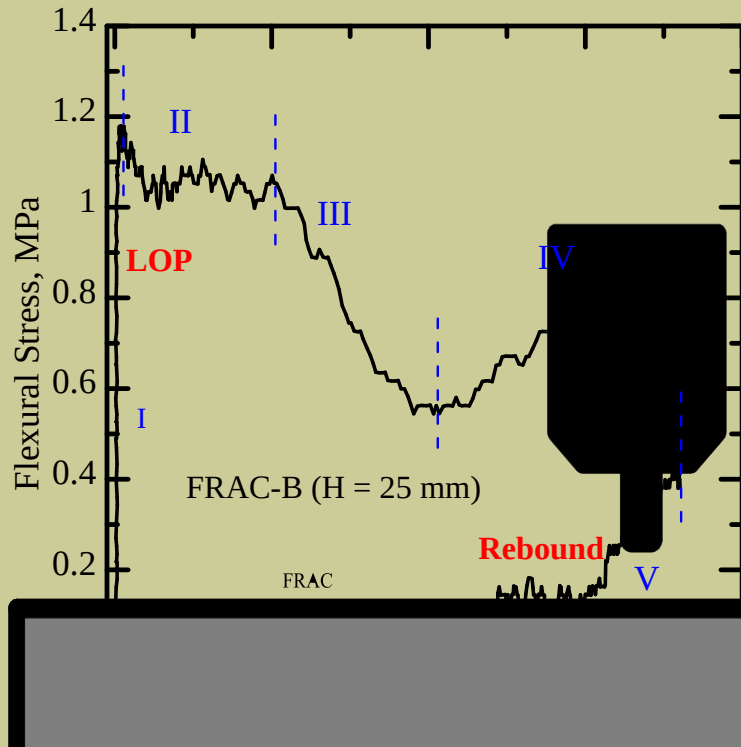
Fiber Pullout

Impact Test - Instrumentation

- ❑ Electronic hoist and release mechanism to trigger the impactor
- ❑ Pneumatic brake system using a contact type micro switch
- ❑ Experimental setup includes: Signal conditioning amplifiers, National Instruments PCI acquisition system, LABVIEW VI with a trigger function
- ❑ Data recorded at a sampling rate of 20 kHz, Duration of recorded data - 200



Typical Impact Responses - FRAC



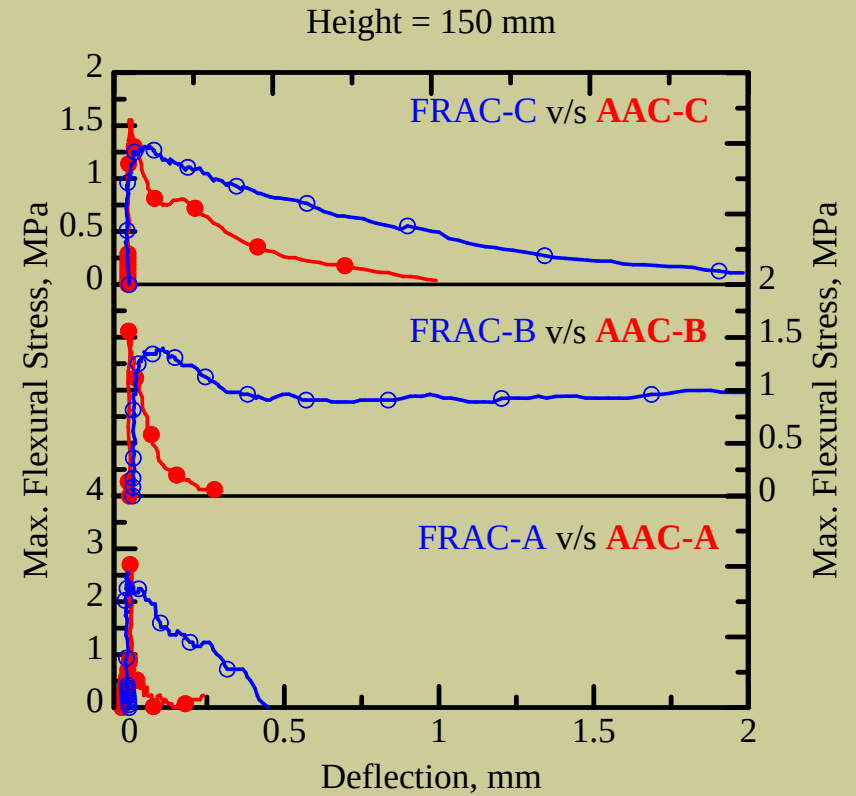
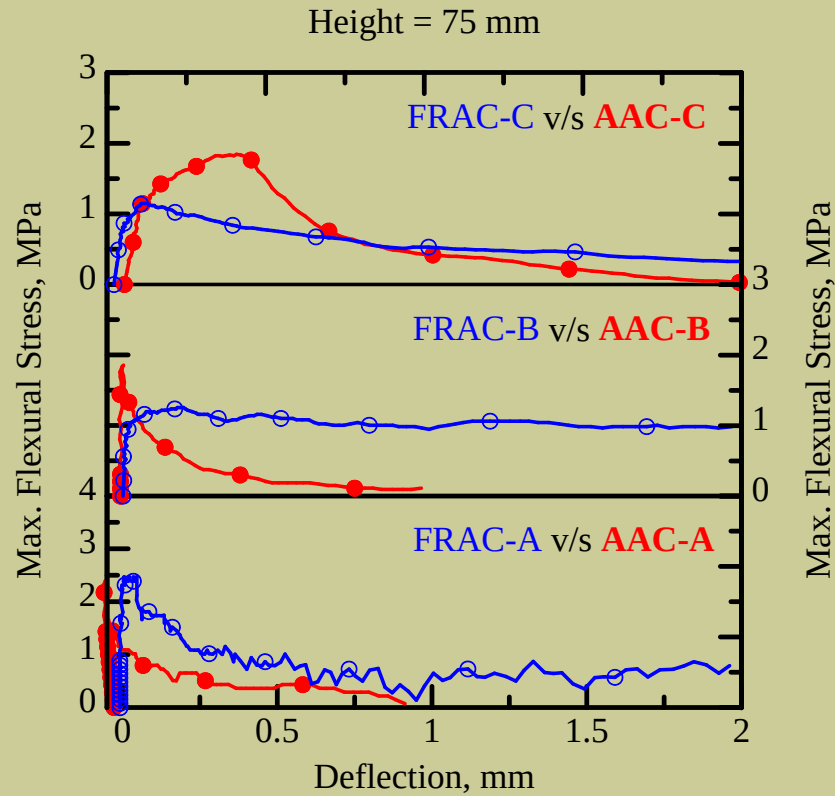
Flexural stress versus deflection response can be categorized as:

Zone I – Linear Elastic Range (Initial Rigidity), ends with first crack

Zone II – Multiple cracks (Strain Softening Mechanism)

Zone III and IV – Stiffness Degradation (Fiber bridging Mechanism)

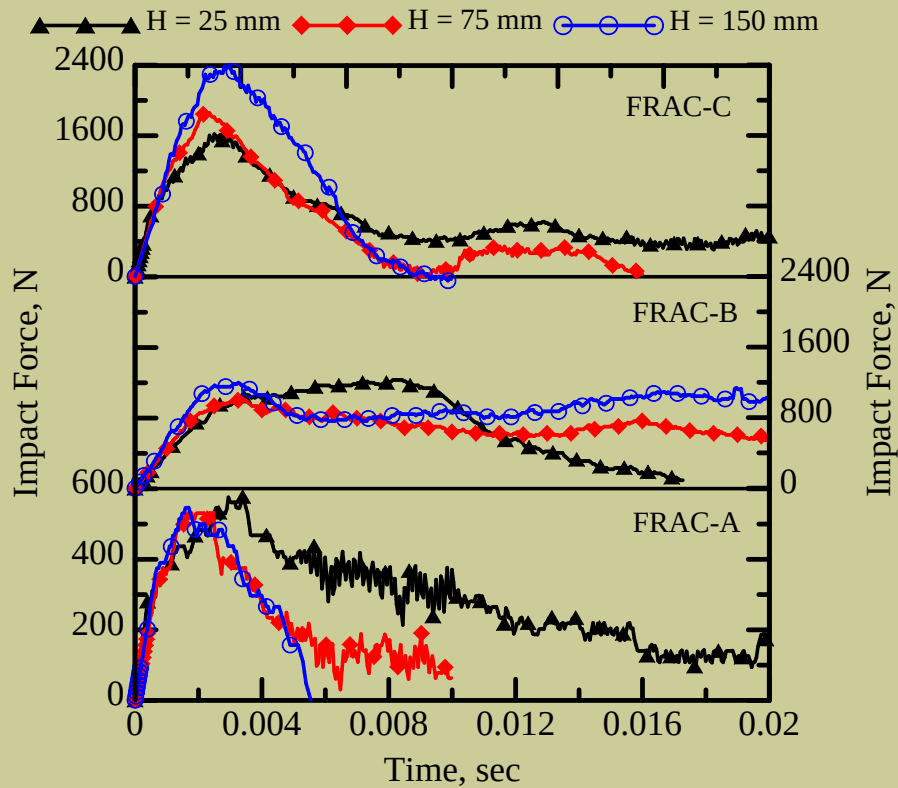
FRAC v/s AAC



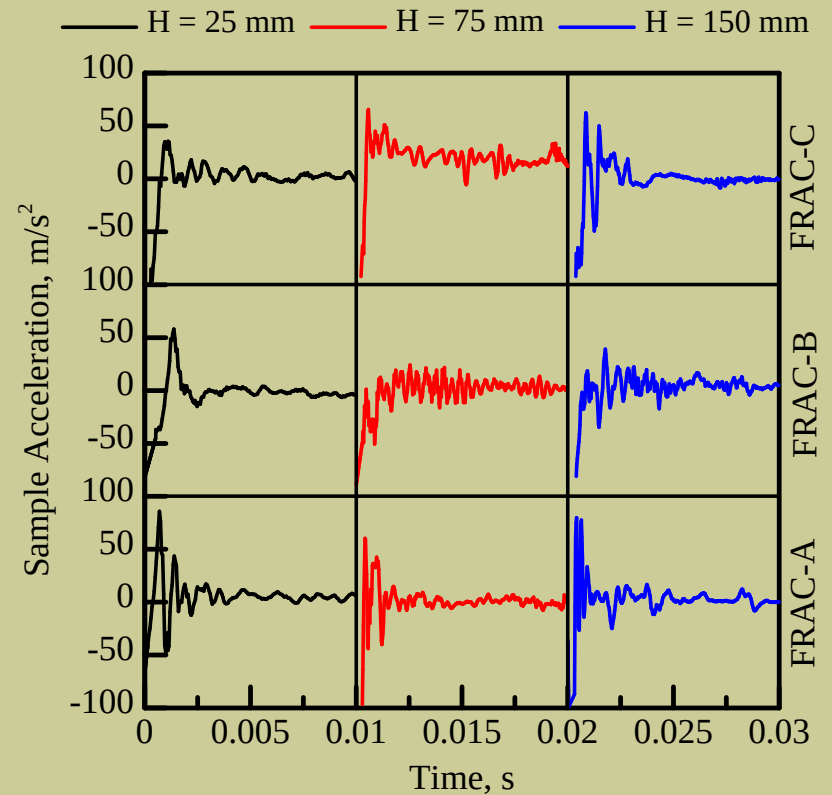
Drop Height = 75 mm

Drop Height = 150 mm

Effect of drop height of impactor

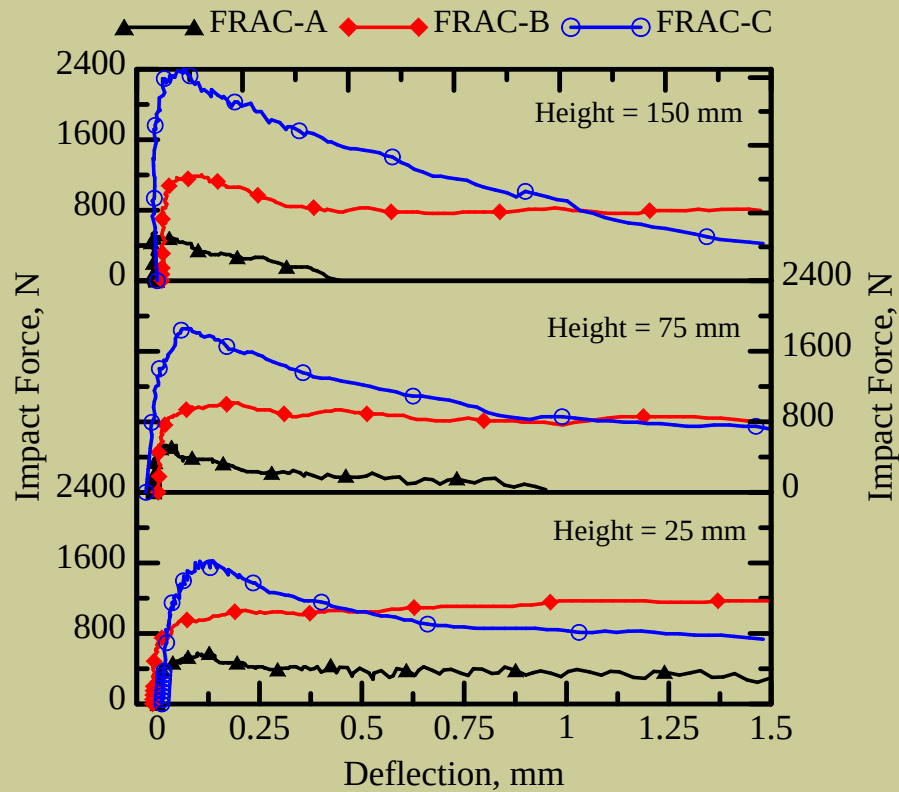


Time History of Impact Force

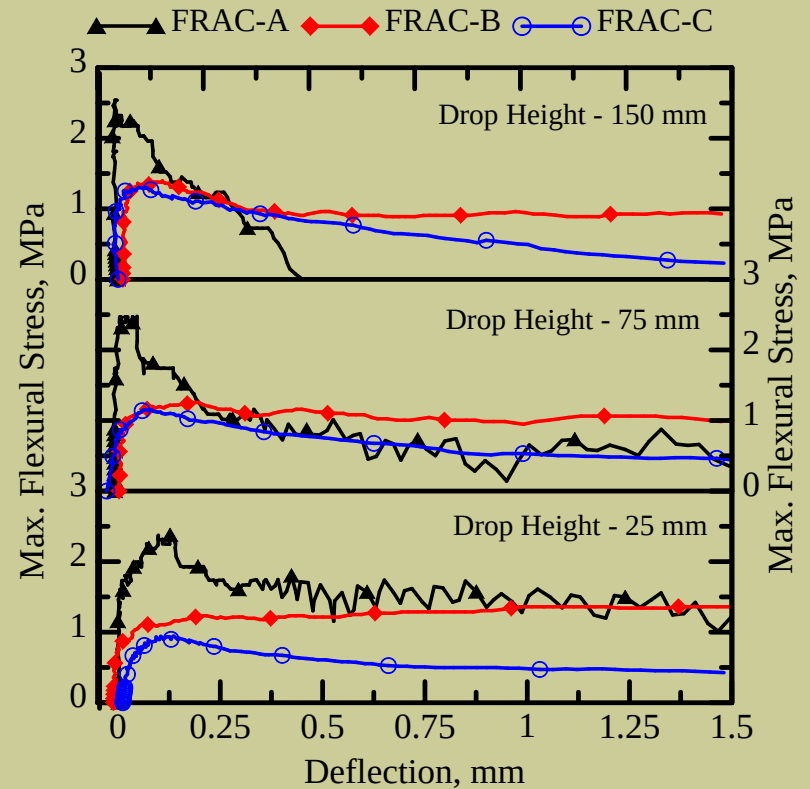


Time History of Specimen Acceleration

Specimen Size Effect

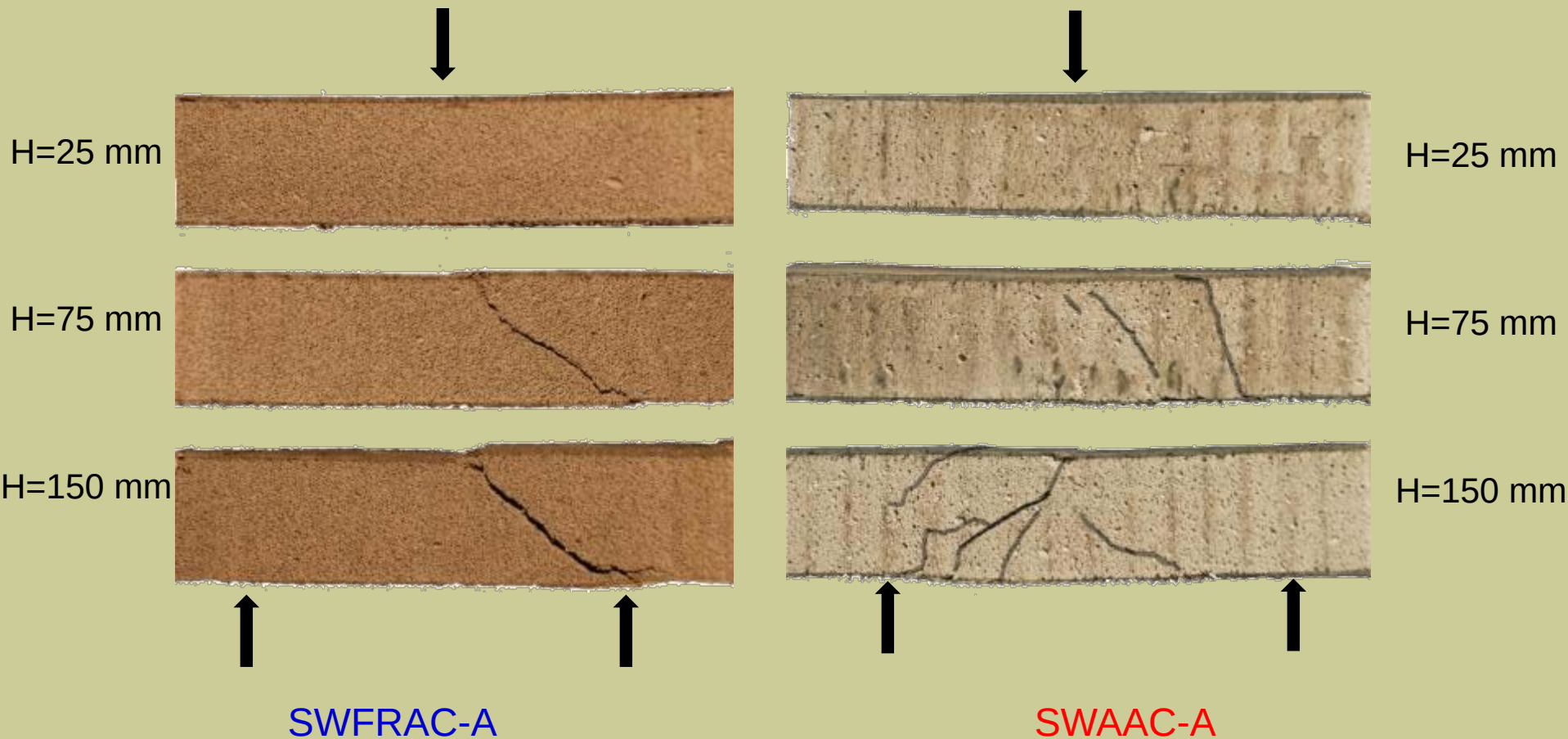


Impact Force v/s Deflection

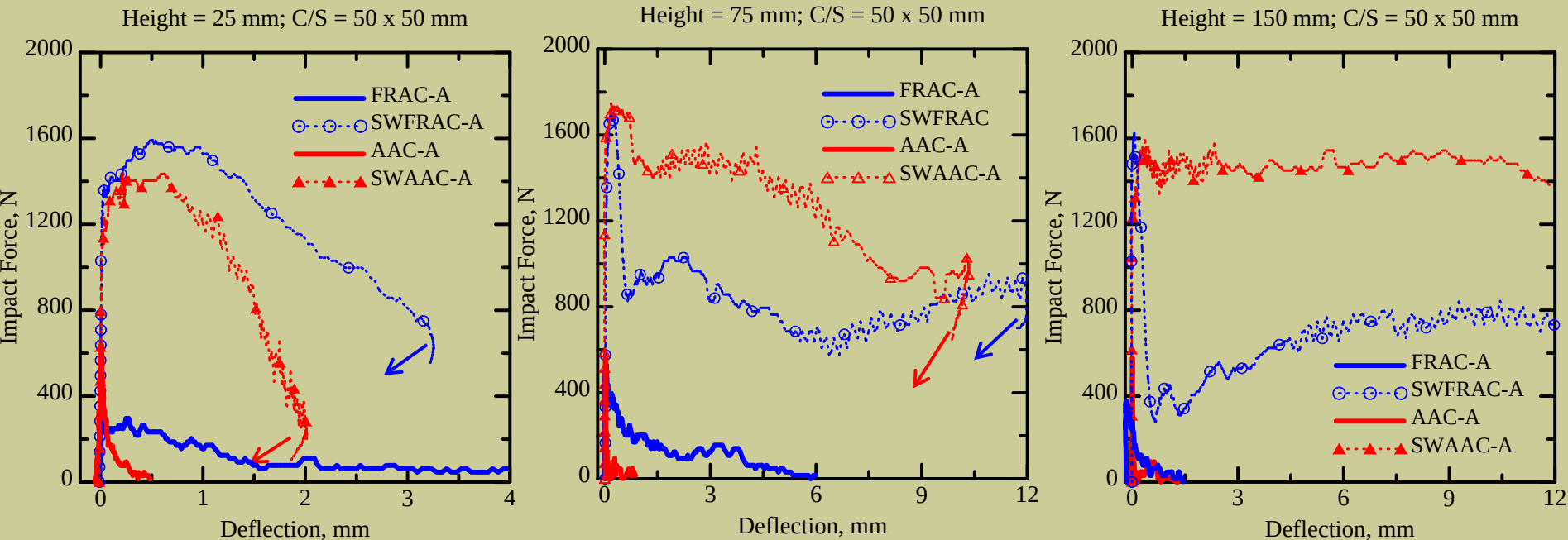


Flexural Stress v/s Deflection

Impact Test on Sandwich Composites



The Sandwich Effect



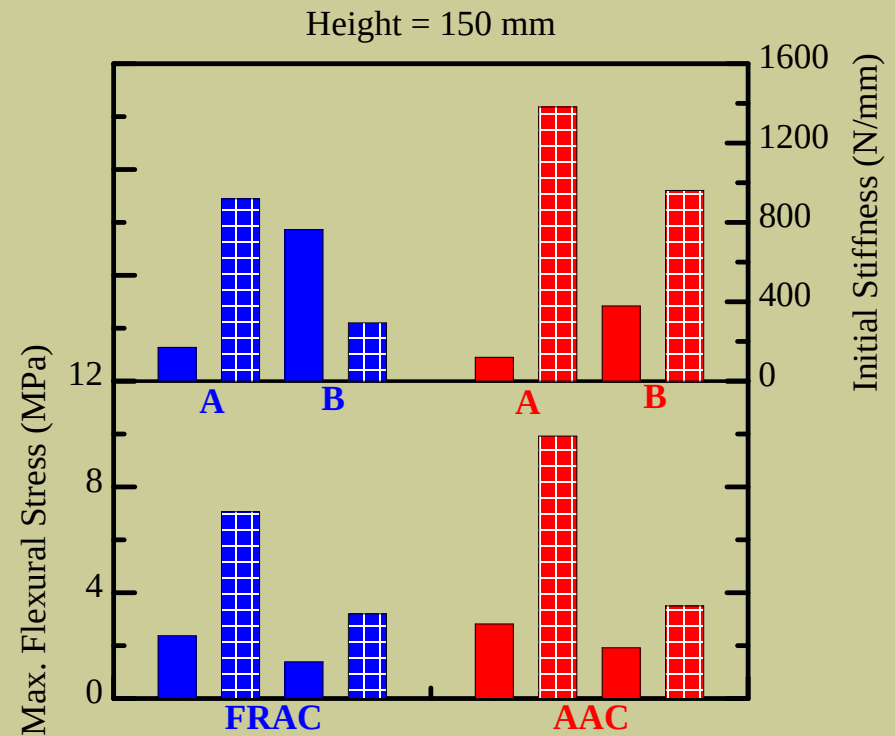
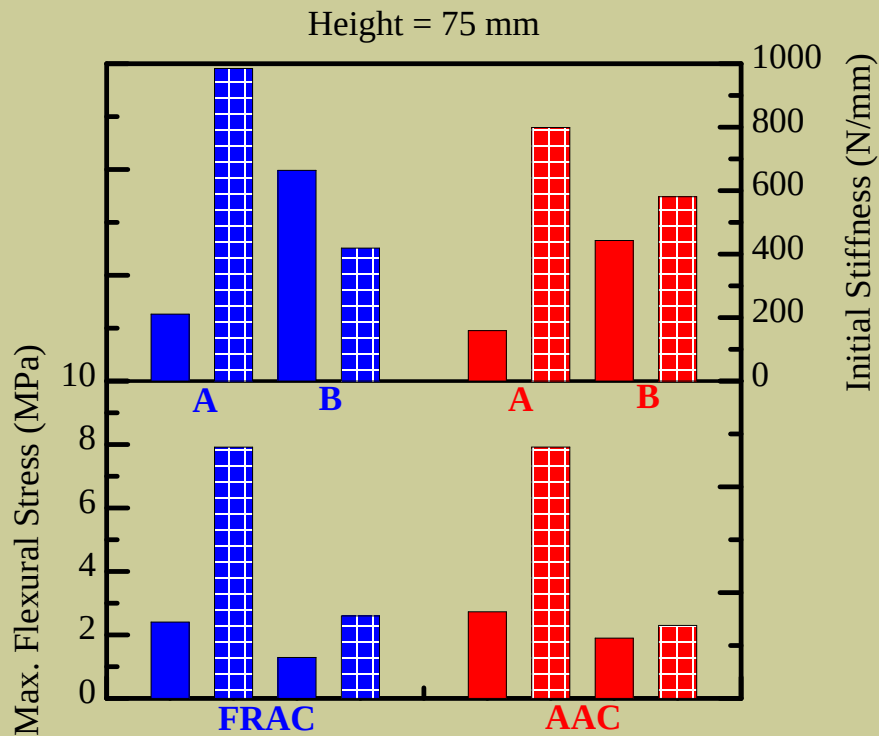
Height = 25 mm

Height = 75 mm

Height = 150 mm

- Effect of textile reinforcement on AC - 3 times increase in peak load and flexural strength and 5 times increase in initial stiffness.
- Energy absorption capacity increases by about 8 times for SWFRAC, and 30 times for SWAAC.

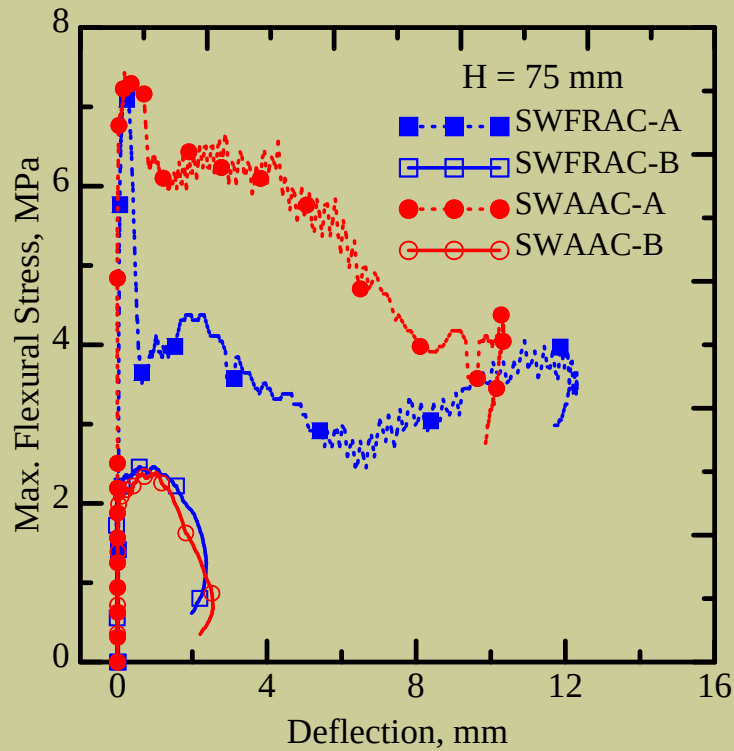
Sandwich Effect - Continued



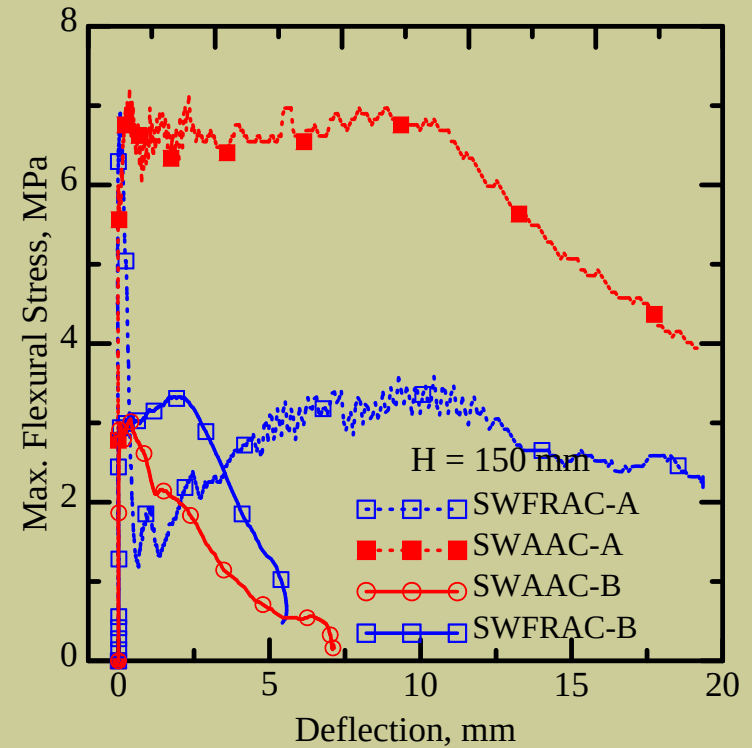
Drop Height = 75 mm

Drop Height = 150 mm

Specimen Size Effect

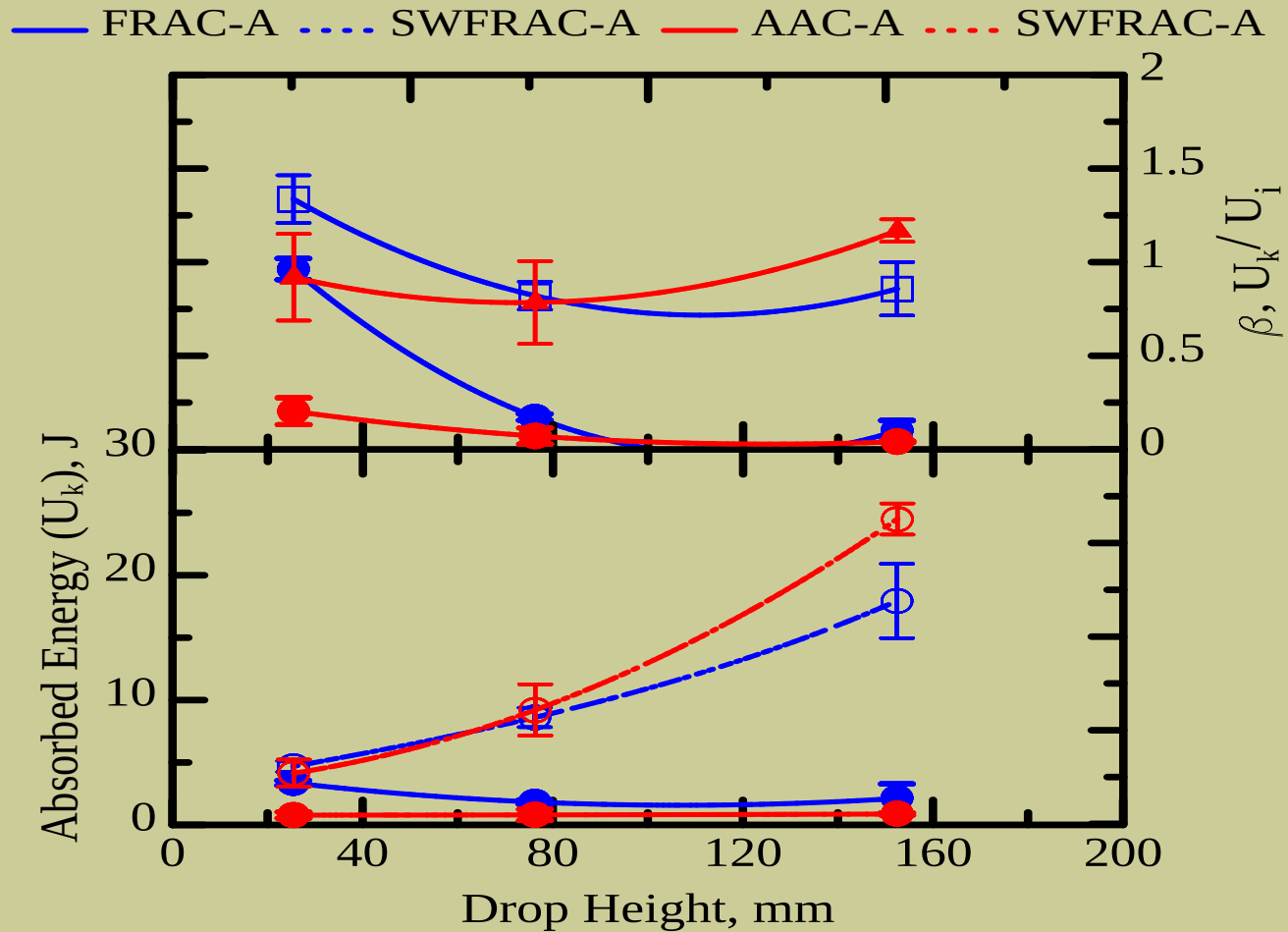


Drop Height = 75 mm



Drop Height = 150 mm

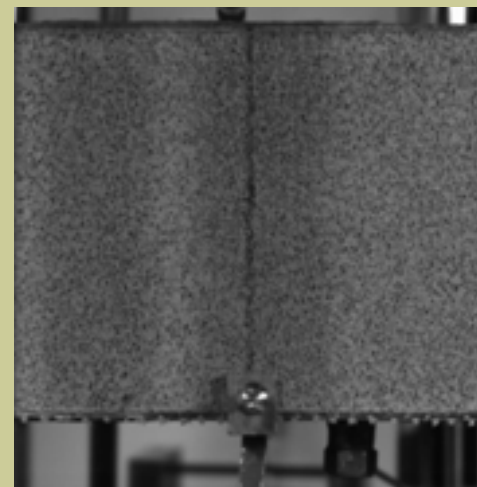
Energy Absorption



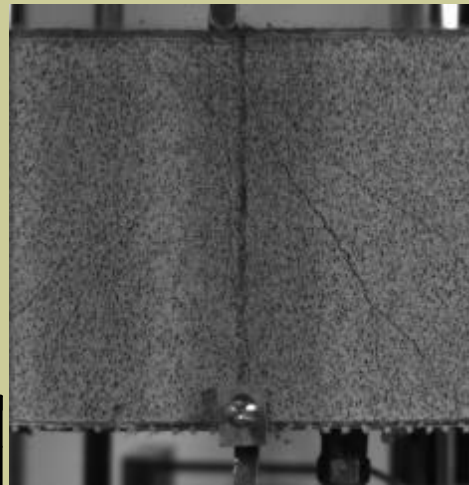
Core (Solid) – **FRAC** & **AAC**

Sandwich (Broken) – **SWFRAC** & **SWAAC**

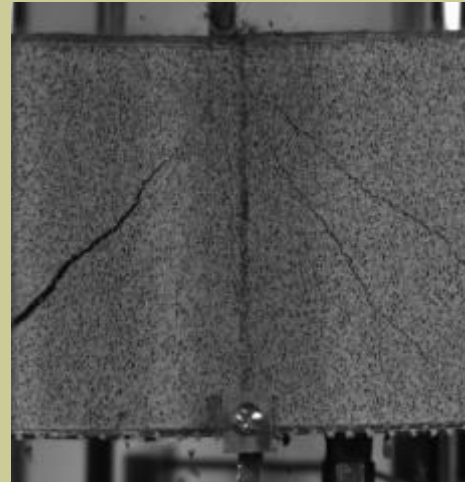
Crack propagation under impact load



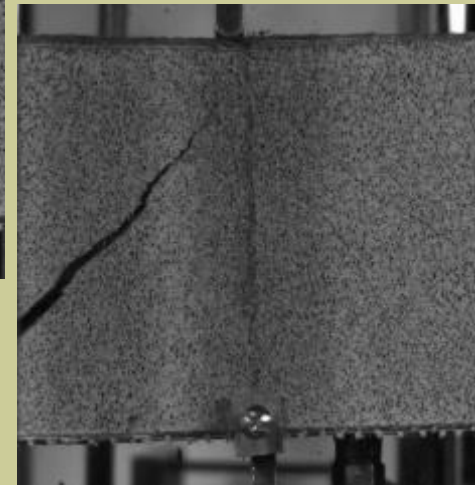
Frame 1



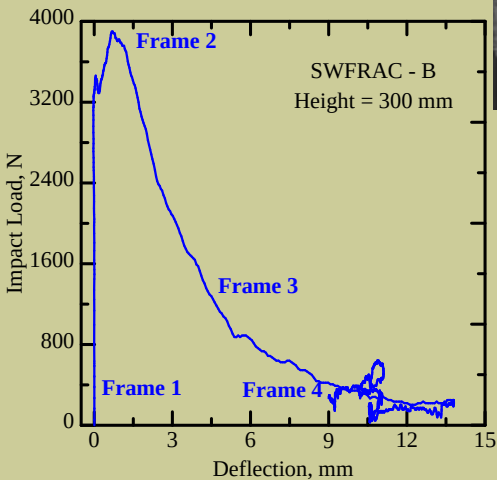
Frame 2



Frame 3



Frame 4



	Frame 1	Frame 2	Frame 3	Frame 4
Time (μsec)	0	6590	11020	70740
Load (N)	0	3899	1295	187
Deflection (mm)	0	0.71	4.43	9.23
Crack Width (mm)	0	0.60	2.41	4.88

Conclusions

- Static flexural and high strain-impact tests were conducted on plain aerated concrete and sandwich insulated panels
- Fiber-reinforced aerated concrete can be considered as a green construction material with low cost of production and good thermal characteristics.
- Sandwich composites reinforced with textiles improves the mechanical properties such as toughness, deformation capacity and flexural strength.
- Sandwich composites were found to be suitable for application as building envelopes in infrastructure systems

	Strength	Ductility
TRC Sandwich	✓✓✓	✓✓✓
Plain AAC	✓	
Plain FRAC		✓